



Improved Mask R-CNN algorithm: Multi-ore detection and positioning based multi-sensor fusion in complex field environment

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ABSTRACT

In metal ore mining, detecting and locating large ore targets for secondary blasting is challenging due to adverse conditions such as high dust levels, dense ore distributions, and low edge contrasts. This paper introduces a multi-ore target detection and positioning method that combines monocular vision with LiDAR technology. Integrating LiDAR distance data with industrial camera pixel information and applying monocular vision principles, we achieve precise detection and positioning of multiple ore targets. The RGB images captured by the industrial camera are processed using an improved Mask R-CNN (Mask Regional Convolutional Neural Network) algorithm. The detection performance of the proposed method was validated using a self-constructed dataset. The experimental results demonstrate that the MioU (mean Intersection over Union) and mAP (mean Average Precision) for multi-ore targets reached 84.9% and 91.7%, respectively, surpassing the original algorithm's performance of 76.0% and 81.7%. The relative error in the lateral distance positioning of multi-ore targets is 3.52%. This method can more effectively accomplish real-time ore target detection and positioning in complex field environment.

1. Introduction

In the metal mining industry, a significant portion of the mining techniques is dedicated to the extraction process within mines. After the initial blasting, a second round of blasting is often necessary to effectively control the size of the remaining large ore fragments. Consequently, the implementation of accurate detection and localization technologies for ore targets is highly valued in ore crushing projects. On construction sites, distinguishing and identifying multiple ore targets within a complex and dynamic environment is crucial to ensure the precision of crushing operations and to obtain real-time location information of these targets. Industrial cameras have become common tools for target detection and segmentation due to their cost-effectiveness, rich visual data, and ease of target identification and classification. However, camera-based detection and segmentation techniques are prone to interference from operating conditions such as intense lighting and high dust concentrations, which can impede the accurate acquisition of ore target location data. LiDAR technology plays a critical role in measuring the relative distance between obstacles and the system in specific scenarios. The laser beams emitted by LiDAR can effectively penetrate dust, but the technology has limitations in capturing the

geometric and classification features of ore targets. To overcome these limitations, an integrated multi-sensor data fusion approach has been adopted to achieve precise segmentation and localization of ore targets. This method combines the strengths of both radar and vision systems: using LiDAR to obtain the relative distance information of ore targets and industrial cameras to collect classification and geometric data of the targets. This synergy enables the precise segmentation and localization of ore targets even in complex and variable operating environments, providing robust support for the accurate operation of crushing hammers.

To advance the intelligent development of mining operations and to facilitate the automated shattering of sizable ores in open-pit mining, a crucial first step is the swift and precise automated recognition of ore targets. Traditional ore crushing operations rely heavily on manual labor, which not only involves significant labor intensity but also poses numerous safety hazards. To address these challenges, extensive research has been conducted, leading to the development of an innovative ore identification technology based on image segmentation algorithms. This cutting-edge technology enables the rapid detection and precise localization of ore targets through accurate image analysis and processing, significantly enhancing the efficiency and safety of ore

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crushing operations. Due to its superior performance and broad applicability, this technology has been widely adopted across numerous mining sites.

Image segmentation stands as a cornerstone issue within the domain of computer vision, integral to a plethora of systems aimed at interpreting visual data. This process is pivotal for segmenting images or video frames into discrete segments and objects. It can be conceptualized as classifying individual pixels with semantic labels (semantic segmentation) [1], delineating separate instances of objects (instance segmentation) [2], or a combination of both (panoptic segmentation) [3]. Instance segmentation builds upon semantic segmentation by not only identifying but also delineating each object of interest within an image. A plethora of image segmentation methodologies have been put forth in scholarly literature. The spectrum of these methods extends from foundational techniques such as thresholding [4], histogram-based segmentation [5], region growth [6], K-means clustering, and watershed algorithms, to more sophisticated approaches like active contours, graph cuts, conditional random fields or Markov random fields, and sparsity-driven techniques. The advent of convolutional neural networks [7] in image processing marks a significant technological leap for handling complex imagery [8–14].

Machine vision technology is revolutionizing the mining and crushing industries, making the study of ore image segmentation and recognition a top research focus. Advancements in image processing have placed these studies at the forefront of innovation, where they are leading the way in developing intelligent solutions for ore detection and analysis. Current methodologies for segmenting ore images are broadly categorized into two types: traditional and those based on deep learning. When employing traditional segmentation techniques to address intricate ore imagery, several challenges emerge. For instance, the co-representation and bipartite graph model-based segmentation approach detailed in Reference [15], while capable of creating a sparse matrix from bipartite graphs and utilizing a graph cut technique in conjunction with clustering for segmentation, struggles with the regulation of cluster quantities. Reference [16] explores hardware-accelerated methods to enhance segmentation efficacy; however, these require ongoing fine-tuning to align with varying operational contexts. In Reference [17], a Harris corner detection technique predicated on bilateral filtering is introduced. This method delineates boundaries to mitigate merging by aligning the distance of concave points with the angle's extremities. Nonetheless, it necessitates the customization of corner point parameters to suit diverse scenarios. Reference [18] amalgamates watershed segmentation markers with shaded areas by selecting a threshold-based gradient to isolate the minimum point. Yet, this approach is not optimized for the concurrent detection of ores varying in dimensions. Traditional segmentation strategies, which rely on the color and texture of the imagery, are prone to inaccuracies in ore edge delineation. This is attributable to the intricate texture of ores and the close color similarity between the ore targets and their backdrops, which can confound segmentation processes [19].

Deep learning-based segmentation algorithms harness the power of convolutional neural networks (CNNs) to extract the features of ore targets, thereby facilitating segmentation. Over the past few years, the field of deep learning has seen exponential growth and has been extensively applied to a multitude of image processing tasks. It has also made inroads into the domain of intelligent ore sorting, where deep learning is being leveraged to address challenges in ore image segmentation and recognition. Studies have indicated that deep learning algorithms offer enhanced speed and precision. In Reference [20], the U-Net architecture was introduced for the segmentation of seabed ore images, demonstrating high accuracy in segmentation. This model has been lauded for its effectiveness. In Reference [21], the UR model was introduced for automated image analysis in mining engineering, showing its innovative approach and high effectiveness. Meanwhile, Reference [22] suggests utilizing a nested edge detection model that builds upon the VGG network for segmentation tasks. This approach

shows promise in identifying larger ore bodies effectively, yet it falls short when it comes to the recognition of smaller ore fragments. In Reference [23] presents an adaptation of the DUNet model, which achieves segmentation tasks by adaptively adjusting the receptive field to capture the edges of ores. In Reference [24] introduces the LosNet framework, enabling efficient segmentation of ore images specifically tailored for environments with constrained resources. In Reference [25] validates the utility of convolutional neural network-based semantic segmentation in the mineralogical analysis of sandstone micro images, offering an effective classification framework for 10 distinct rock categories with notably high predictive accuracy. In Reference [26] presents an advanced semantic segmentation method designed to accurately identify transparent minerals within complex geological materials, thereby providing an efficient technical strategy for advancing mineralogical analysis. In Reference [27] designed based on Mask R-CNN, the lightweight network significantly improves the detection accuracy and speed of coal gangue sorting and has been successfully applied in complex scenarios.

While the traditional methods previously discussed may yield some results, their efficacy in recognition is suboptimal, particularly when dealing with dense ore target distributions and low contrast edges in images from field blasting operations. These methods often fall short in achieving the segmentation precision demanded for identifying ores in the blasted heaps of open-pit mines. Conversely, the deep learning approach, characterized by its swift recognition capabilities and elevated segmentation accuracy, is predominantly tailored for ore images that feature consistent backgrounds. This makes it less versatile when applied to the analysis of blast ore images within the intricate and variable conditions typically encountered in open-pit mining environments.

This manuscript presents an enhancement of the Mask R-CNN (Region-based Convolutional Neural Network with a Segmentation Mask) instance segmentation framework. The improvement involves refining the network's feature extraction capabilities and integrating a bidirectional feature pyramid network. This modification addresses the challenge of inaccurate edge detection in the segmentation of multiple ore targets, enabling the precise detection of multiple ore targets. Subsequently, the pixel coordinate data captured by an industrial camera are combined with the lidar's target distance measurements. By applying the principles of monocular vision positioning, the three-dimensional spatial coordinates of the ore are ascertained, thereby achieving accurate ore localization. The experimental results demonstrate a notable enhancement in detection precision with this approach. It allows for real-time ore detection and significantly diminishes the relative positioning error of ore distances when leveraging multi-sensor information fusion, as opposed to relying on a single sensor. These outcomes substantiate the method's efficacy. The primary contributions of this study can be summarized as follows:

- Optimizes feature extraction and incorporates a bidirectional feature pyramid network, resulting in higher mIoU and mAP scores, outperforming traditional segmentation methods.
- Integrates LiDAR distance data with monocular vision for precise multi-ore target detection and localization, significantly enhancing accuracy in complex mining environments.
- Utilizes monocular vision principles to accurately determine the three-dimensional spatial coordinates of ore targets, achieving a relative lateral distance error of just 3.52 %, ensuring precise and reliable ore localization for secondary blasting operations.

The rest portion of the paper is organized as follows: Section 2 introduces the fundamental principles of Mask R-CNN and the corresponding improvement methods, and Section 3 provides a detailed description of the multi-ore target feature information fusion technology. We analyze the experimental results in Section 4 and finally, Section 5 presents the conclusions.

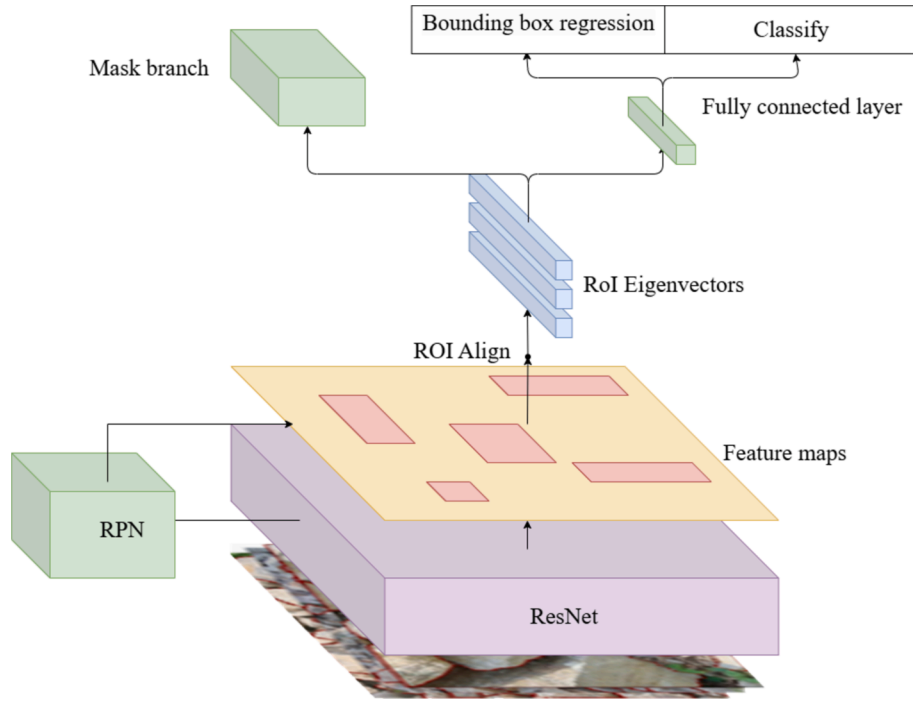


Fig. 1. Mask R-CNN ore detection segmentation structure diagram.

2. The basic principle of Mask R-CNN

2.1. The basic architecture of Mask R-CNN

In traditional image segmentation research, network architectures have often focused on the precise identification and segmentation of single targets. For the image dataset addressed in this project, characterized by the presence of multiple large rock formations, it is necessary not only to accurately localize the bounding boxes for each rock but also to achieve detailed segmentation of the respective rock areas. Given the complexity of the scenes, standard image segmentation methods may face performance limitations and may not meet the project's stringent

requirements for precision and detail. This study will implement ore detection and segmentation based on Mask R-CNN [28], the structure of Mask R-CNN is shown in the Fig. 1. The structure uses the residual network ResNet50 as the backbone network, and the FPN is used to extract the stones, and the execution process of the Mask R-CNN algorithm for the stone instance segmentation network is as follows: firstly, the input stone image data is fixed to the specified size and input into the pre-trained ResNet50 + FPN network model, the features are extracted and the corresponding feature maps are obtained, and a large number of RoI is obtained after the feature maps pass the target detector RPN, Then, with the help of Softmax classifier, the foreground and background were classified binarily, the bounding box regression was used to

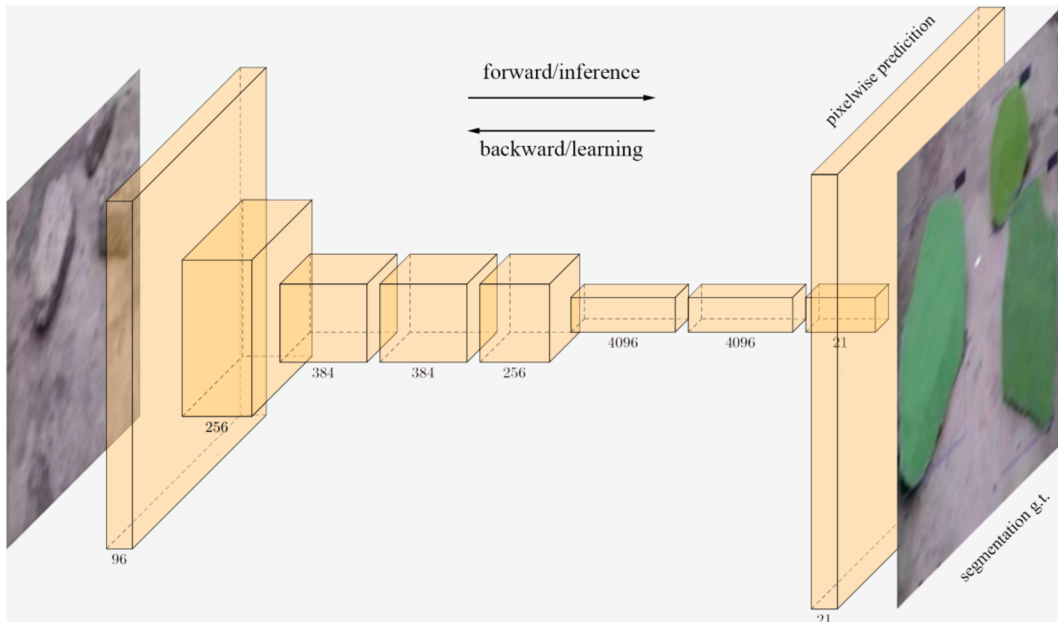


Fig. 2. Schematic diagram of FCN structure.

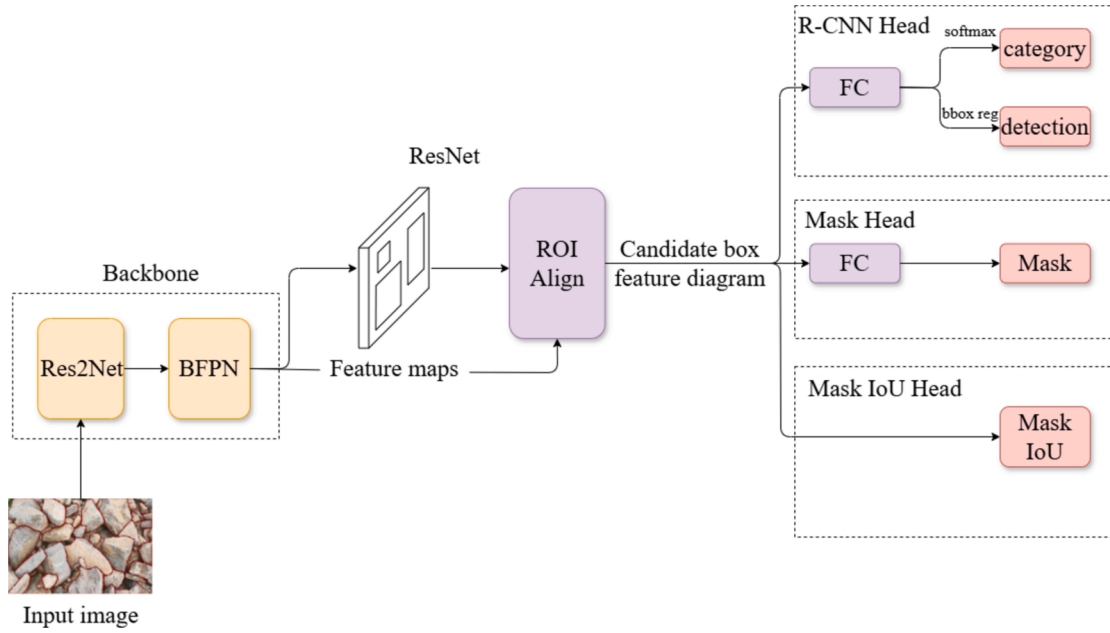


Fig. 3. The framework diagram of the Mask R-CNN network structure has been improved.

obtain more accurate candidate box position information, and part of the RoI was filtered out by non-maximal suppression. Send the feature map and the remaining RoI to RoIAlign for size unification to generate a fixed-size feature map; The feature map with uniform size was passed through a double-branch detection head, one of which used a fully connected layer for target classification and bounding box regression, and the other class used FCN for pixel-level segmentation.

The introduction of the Fully Convolutional Network (FCN) is aimed at meeting the high precision requirements of the output information in Mask R-CNN. By decoupling the tasks of category classification and boundary regression, it significantly enhances the recognition accuracy of object instances and achieves pixel-level mask extraction for regions of interest. As an advanced semantic segmentation algorithm, FCN classifies at the pixel level, generating fine binary masks for each object instance. As shown in Fig. 2, the structural design of FCN ensures the accuracy of the entire process from image compression, predictive classification to upsampling restoration, thereby achieving clear segmentation of each instance in the stone image at the pixel level. The

advantage of this method is that it can provide richer and more accurate visual information for complex image content, especially when dealing with multi-object and complex background scenarios, demonstrating excellent segmentation performance.

2.2. Enhancing the Mask R-CNN framework

2.2.1. Improve the basic framework of the network

This study employs Res2Net101 as the backbone network, integrated with the Bidirectional Feature Pyramid Network (BFPN), to facilitate the extraction of multi-scale features, thereby enhancing the model's ability to recognize targets of varying sizes. Utilizing ROI Align, we map the Region of Interest (ROI) proposals generated by the Region Proposal Network (RPN) onto a consistent feature map, ensuring the precision of feature extraction. Subsequently, in the detection branch of the model, we perform object classification and bounding box regression on the feature maps, while in the segmentation branch, we introduce the Mask IoU Head for optimization to produce accurate segmentation maps.

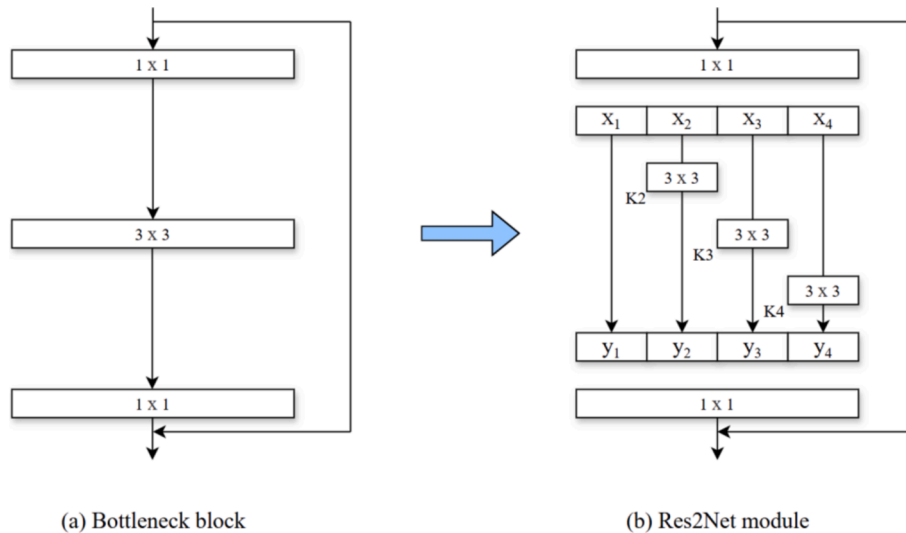


Fig. 4. Res2Net Fabric Network.

Fig. 3 succinctly displays the structure of the improved Mask R-CNN.

2.2.2. Optimize the design of feature extraction network

Multi-scale feature extraction is of great significance in computer vision, and designing a more effective structure can improve the ability of the network to extract features. In the process of ore identification, due to the high density of ores, the occlusion between ores is serious, which can easily lead to false detection and missed detection, so more abundant information is needed to represent the characteristics of ores. Res2Net, an improved version of ResNet, constructs hierarchical residual connections within individual residual blocks, enhancing the receptive field and obtaining multi-scale feature representation. This optimization allows the network to more comprehensively capture local details and global context in ore images, significantly improving the expressive power of ore features. The network output can be expressed by the following formula, as shown in the following Fig. 4. On the left is the basic network module of ResNet, and on the right is the structure of Res2Net. Firstly, the feature map of the 1×1 convolution output is divided into 4 (s) modules, in which x_1 is directly passed to y_1 , x_2 is passed into y_2 after 3×3 convolution, x_3 is fused with y_2 and 3×3 convolution and output as y_3 , x_4 is also fused with y_3 and 3×3 convolution and output as y_4 , and finally the four outputs are fused and passed into the 1×1 convolution.

$$y_i = \begin{cases} x_i i = 1 \\ k_i(x_i + y_{i-1}) 1 < i \leq s \end{cases} \quad (1)$$

The Res2Net module replaces the large 3×3 filters in the bottleneck block of the residual block with multiple small 3×3 filters and connects the small filters through a hierarchical transfer. Each small 3×3 convolution receives all the convolutions on its left except x_1 , the feature layer is constantly fused, and each output can increase the receptive field, the s in the above equation is the control dimension, the larger the parameter s , the more modules are divided, and more features are allowed to be learned with richer receptive field size, so each module can get a different number of feature combinations with receptive fields, to achieve a more efficient feature processing speed.

Res2Net represents multi-scale features with a high degree of granularity and expands the receptive field of each layer, enabling it to capture multi-scale features. To increase the receptive field while reducing the number of layers as much as possible, the 3×3 convolution of x_1 is omitted, which can be seen as a reuse of features. In the substitution process, the hierarchical residual connection does not increase the number of parameters, so it does not increase the burden on the network.

2.2.3. Superimposed bidirectional feature pyramid network design

In deep neural networks, shallow features focus more on detailed information, while higher-level information focuses more on semantic information. Traditional FPN fuses high-level information upsampling with low-level features, which is essentially limited by one-way information flow. Only the high-level semantic information is added, while the shallow semantic information of the neural network is highly correlated with detailed features such as edge shapes. The ore in this paper seriously adheres to the ore, and the extraction of its characteristic parameters requires precise edge detection. To fuse more features, we introduce a Bidirectional Fusion Feature Pyramid Network (BFPN), an innovative improvement over FPN. BFPN incorporates a dual information flow mechanism with multiple overlays, retaining FPN's bottom-up flow and adding a top-down downsampling flow. This effectively fuses low-level edge features with high-level semantic features, enhancing sensitivity to edge details and improving the precision of ore edge feature extraction. BFPN's framework enriches feature representation, enabling comprehensive capture of complex structures and subtle features in ore images. This innovative structure provides robust support for accurate segmentation and feature extraction of ore images,

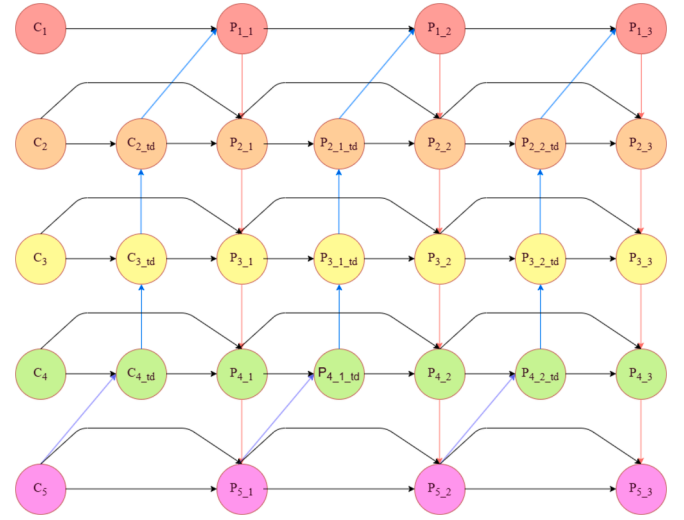


Fig. 5. BFPN Fabric Network.

overcoming the lack of detail in deep feature layers of traditional FPNs and improving recognition accuracy in complex scenes. The BFPN characteristic pyramid network is shown in the following Fig. 5. C_1 - C_5 are the output feature maps of Res2Net101, which are used as the input of the feature pyramid, the downward arrow indicates that the feature map is downsampled by the convolution with a stride of 2, and the upward arrow indicates the upsampling operation.

Firstly, the nodes with only one output edge without feature fusion in the process of upsampling and shallow feature fusion of the original FPN are removed to reduce the resource consumption of nodes with small contributions to feature fusion. Secondly, when the shallow features are fused with the high-level features through downsampling, an additional edge with high-level information at the same level as the output node (such as the generation of the feature layer P_4_1 , in addition to the fusion of the P_3_1 downsampled information and the horizontal links of the C_4_{td} , C_4 with rich semantic features is introduced) to fuse more features. Finally, considering the limited information mentioned in a single fusion, to further mine more features, each bidirectional path is regarded as a one-layer feature network, and the same feature network is stacked three times to obtain a higher level of scale fusion.

3. Multi-ore target feature information fusion

The industrial camera captures RGB images through image sampling and inputs them into the trained network model to obtain ore segmentation results, and the lidar obtains the output differential signal through the process of transmitting waves and receiving waves to obtain the distance information of obstacles and the pixel coordinate information of the imaging plane of the ore target for data fusion and obtains the three-dimensional spatial coordinates of the ore target based on the monocular positioning principle to realize the detection and positioning of the ore target, and its principle block diagram is shown below.

3.1. Ore target matching

The principle of monocular vision positioning using an industrial camera involves capturing a single frame image of the ore target. This setup forms a triangle comprising the plane imaging point, the actual three-dimensional space point, the lens light point, and a point on the same horizontal plane. By applying the principle of triangle similarity, the three-dimensional spatial coordinates of the ore target object can be determined. The model illustrating monocular vision with an industrial camera is shown in the Fig. 7 below.

At any point in the range of the industrial camera lens, P passes

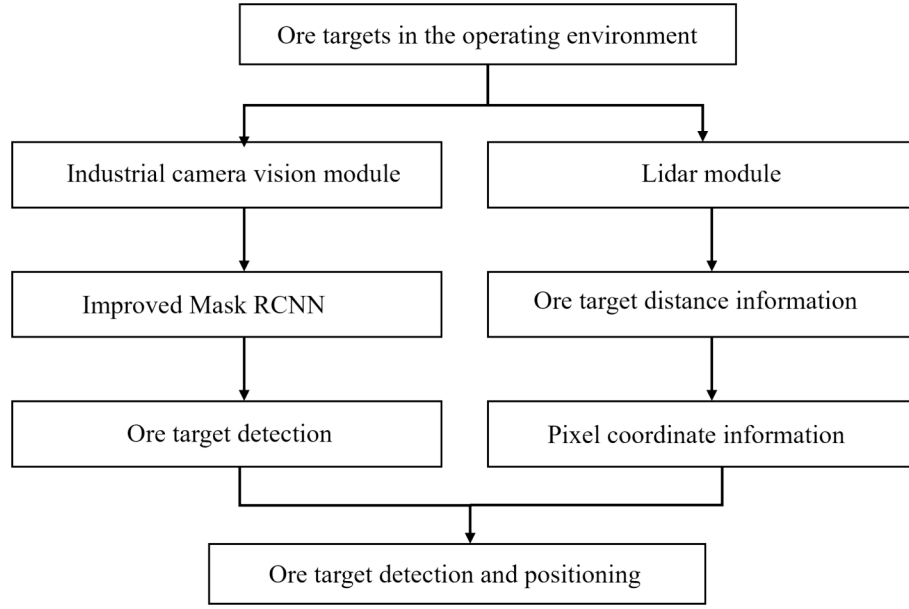


Fig. 6. Schematic diagram of ore segmentation and positioning.

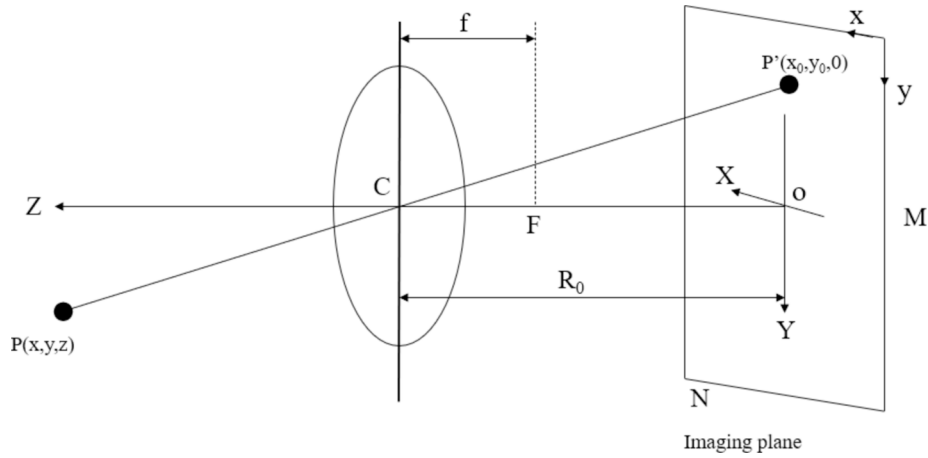


Fig. 7. Monocular vision imaging model diagram of an industrial camera.

through the industrial camera lens and appears in the imaging plane as P' . The focal length of the industrial camera lens is f , the intersection of the imaging plane and the optical axis is the coordinate origin O , the right side of the imaging plane is the X axis, the upper is the Y axis, and the optical axis is the Z axis to establish the imaging coordinate system O - XYZ . The coordinate system o - xy in the imaging plane is used to describe the coordinates of pixels, where the distance between the point O and the optical center C is R_0 .

In the imaging coordinate system O - XYZ , the coordinates of point P are set to $P(x,y,z)$, the image point $P'(x_0,y_0,z_0)$ of point P in the imaging plane, and the coordinates of the optical center are $C(0,0,R_0)$, and the coordinates of the image point P' in the pixel coordinate system are $P'(m,n)$ from the image after image processing, assuming that the size of a pixel is d_0 , where M represents the total number of points of the longitudinal image of the imaging plane, and N represents the total number of points of the horizontal image of the imaging plane, The position of P' in the pixel coordinate system can be calculated as:

$$\begin{cases} x_0 = \left(-\frac{N}{2} + m\right)d_0 \\ y_0 = \left(\frac{M}{2} - n\right)d_0 \end{cases} \quad \# \quad (2)$$

Vector $\overrightarrow{PP'} = (x_0 - x, y_0 - y, -z)$, $\overrightarrow{PC} = (-x, -y, R_0 - z)$, Vector $\overrightarrow{PP'}$ is proportional to the corresponding coordinates of vector $\overrightarrow{PC}$, and the modulus of vector $\overrightarrow{PC}$ represents the distance R of the ore target obstacle, from which the following equation can be obtained:

$$\begin{cases} \frac{-x}{x_0 - x} = \frac{-y}{y_0 - y} = \frac{R_0 - z}{-z} \\ x^2 + y^2 + (R_0 - z)^2 = R^2 \end{cases} \quad \# \quad (3)$$

According to the above equation, it can be obtained:

Table 1
Lidar and industrial camera fusion method.

Characteristic information	Fusion approach
Target type	The segmentation results of industrial cameras based on the improved Mask R-CNN algorithm are adopted
Target width	It is calculated according to the longitudinal distance detected by the lidar and the angle of the edge of the ore detected by the industrial camera
Lateral distance	According to LiDAR, it can be known
Longitudinal distance	According to LiDAR, it can be known
Relative velocity	According to LiDAR, it can be known

$$\begin{cases} x = \frac{R}{R - \sqrt{R_0^2 + x_0^2 + y_0^2}} x_0 \\ y = \frac{R}{R - \sqrt{R_0^2 + x_0^2 + y_0^2}} y_0 \quad \# \\ z = \frac{\sqrt{R_0^2 + x_0^2 + y_0^2}}{R - \sqrt{R_0^2 + x_0^2 + y_0^2}} R_0 \end{cases} \quad (4)$$

From the above equation, it can be proved that the spatial three-dimensional coordinates of the coordinate point P' can be obtained by knowing the pixel coordinates of the imaging plane, the distance between the optical center point and the imaging plane, and the vertical optical center distance of the ore target obstacle.

It can be seen that the ore coordinates detected by the industrial camera imaging plane can be converted into the world coordinate system, and the laser radar is installed directly above the rotation center of the hydraulic crusher, that is, the installation position is determined, and the laser radar can be adjusted to the position in front through the above-mentioned calibration method, and the target information can also be converted into the world coordinate system, and the more information detected by the industrial camera can be correlated with the target information detected by the laser radar.

3.2. Sensor feature information fusion method

Because the ore is usually stacked and the reflected energy is small, there is a situation of missing the detection of the ore, and the method of fusing the industrial camera and the lidar by using the tandem model is easy to leads to the problem of missing the detection of the ore target. After the ore target matching is completed, the ore feature information is fused, and the accurate position information of the ore target and the ore movement information caused by the operation process is determined by using the model fusion advantage of the sensor, and the fusion method is shown in the following Table 1.

Suppose the distances of λ ore targets measured by LiDAR are respectively:

$$L1 < L2 < L3 < \dots < L\lambda - 1 < L\lambda \quad \# \quad (5)$$

The relative size of λ ore targets measured with an industrial camera and their corresponding pixel coordinates in the image are:

$$\begin{cases} d1 < d2 < d3 < \dots < d\lambda - 1 < d\lambda \\ (m1, n1), (m2, n2), (m3, n3), \dots, (m\lambda - 1, n\lambda - 1), (m\lambda, n\lambda) \quad \# \end{cases} \quad (6)$$

Then the actual distance of the points in the image with pixel coordinates is. Bringing the results into the above equation can accurately obtain the three-dimensional spatial coordinates of the point P.

3.3. Sensor target fusion strategy

When utilizing LiDAR and industrial cameras to perceive the

Table 2
LiDAR and industrial camera fusion strategy.

Lidar	Industrial cameras	Decision-making results
Disabled	Disabled	No targets detected
Enabled	Disabled	No targets detected
Disabled	Enabled	Ore targets detected; characteristics defined by industrial camera analysis
Enabled	Enabled	Ore targets confirmed; characteristics provided by synergistic sensor analysis

Table 3
MIOU₅₀, MIOU₇₅, MIOU₈₅, and MIOU test results for 4 networks.

Network model	MIOU ₅₀	MIOU ₇₅	MIOU ₈₅	MIOU
Mask R-CNN	0.785	0.759	0.736	0.760
YOLOACT	0.824	0.851	0.796	0.824
U-Net	0.813	0.826	0.832	0.824
Improved Mask R-CNN	0.857	0.862	0.829	0.849

working environment of hydraulic breakers, the inherent differences in information acquisition and processing between the two sensors may lead to inconsistencies in detection results under certain conditions. Therefore, when integrating data from both sensors, it is necessary to consider their complementary characteristics comprehensively to resolve potential conflicts in detection outcomes, thereby achieving more accurate and reliable environmental perception., and the fusion strategy of LiDAR and industrial camera in this paper is shown in the following Table 2. Table 3 displays the individual detection outcomes of industrial cameras and LiDAR when operating independently, along with the integrated decision-making results achieved when both sensors are used simultaneously.

4. Experimental results and analysis

4.1. Experimental platform

Experimental environment: Processor Inter(R) Xeon(R) W-2123CPU@3.60 GHz, 6G memory, graphics card GeForce GTX 1060Ti, Window10, 64-bit operating system. The improved Mask R-CNN algorithm was trained on this hardware platform, and programmed in Python language, and pytorch was selected as the platform for building the Mask R-CNN network. To solve complex computing problems, the parallel computing architecture CUDA is also used, and the acceleration of GPUs is used to improve the speed of network operation and training. Considering the memory limitations of the hardware platform, the batch size was set to 64 to accommodate the conditions of the experiment.

4.2. Experimental dataset construction

The ore image data in complex construction scenarios was captured by Hikvision DS-IPC-B12HV3-IA industrial camera. To improve the robustness of Mask R-CNN for stone target segmentation, the acquired images were stored as RGB images of 567 pixels * 732 pixels, and 1132 images of various types of stones were finally obtained, taking into account the distance, the illumination of different angles, the occlusion of surrounding weeds, and the overlap between stones.

Labelme was used to manually label the large target stones that meet the crushing standards in each image sample, and the minimum actual shape boundary of the stone was used as the ground truth bounding box during labeling. In the case that the human eye can distinguish between them, all the stone targets in the picture that meet the crushing criteria should be labeled as much as possible, and the ambiguous and too-small stone targets should not be labeled, to avoid the impact of being regarded as negative samples and weakening the algorithm's ability to distinguish positive and negative samples because they are not labeled.

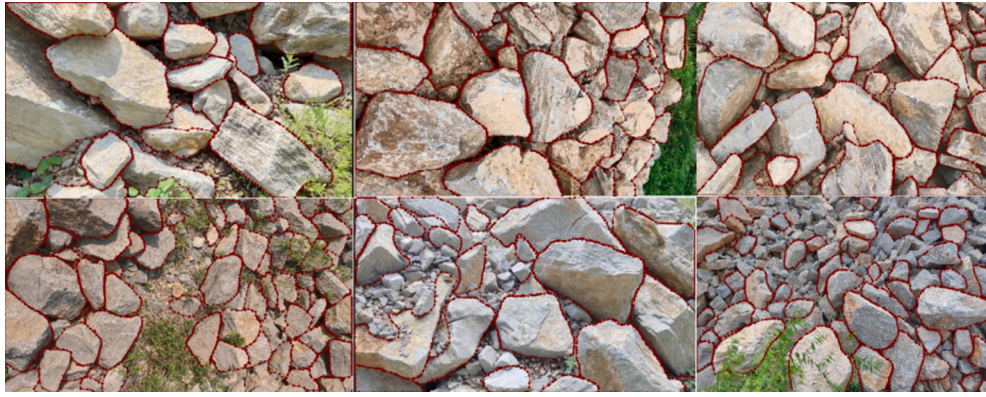


Fig. 8. Manually annotated ore image.

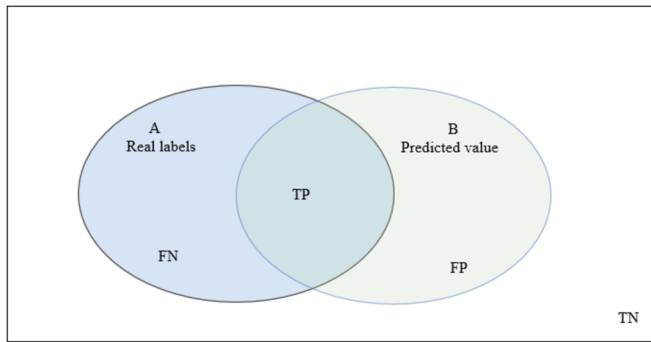


Fig. 9. Schematic diagram of the prediction results.

The image below shows a set of sample images after manual annotation.

During network training, set the network parameters of the pre-trained model to 0.9, set the initial learning rate to 0.01, and set the number of iterations to 500.

Fig. 9 is a schematic diagram of the prediction results, in which TP (True Position) represents the total number of pixels correctly predicted as ore, TN (True Negative) represents the total number of positive sample pixels correctly predicted, FP (False Position) represents the total number of negative sample pixels that are incorrectly predicted as positive samples, and FN (False Negative) indicates the total number of pixels predicted to be non-ore but to be ore.

The indicators used to evaluate the segmentation effect in segmentation algorithms usually include intersection over union (IoU), mean intersection over union (MIoU), pixel accuracy (PA), and class average pixel accuracy (mAP).

IoU represents the ratio of the intersection of the predicted result and the ground truth to the union of both, specifically, it is calculated as the ratio of the area of overlap between the predicted and true values to the area covered by their union:

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (7)$$

MIoU is the average of IoU values belonging to the same category in the image prediction results, and is calculated as follows:

$$MIoU = \frac{1}{\text{class}} \sum_{i=1}^{\text{class}} IoU_i \quad (8)$$

PA indicates the number of correctly predicted pixels in the prediction result of the image as a percentage of the total number of pixels, and is calculated as follows:

$$PA = \frac{TP + TN}{TP + FP + FN + TN} \quad (9)$$

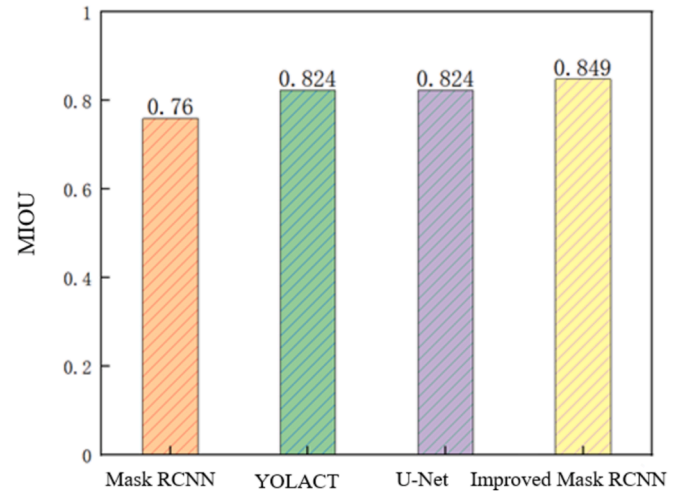


Fig. 10. MIoU test results for 4 networks.

CPA indicates the proportion of correctly predicted images in a single category, that is, the ratio of the number of correctly predicted image pixels to the sum of correctly predicted image pixels and incorrectly predicted non-such image pixels. The calculation formula is as follows:

$$CAP = \frac{TP}{TP + FP} \quad (10)$$

mAP represents the average percentage of correctly predicted pixels for all categories, i.e., the CPAs of all categories are added and averaged:

$$mAP = \frac{1}{\text{class}} \sum_{i=1}^{\text{class}} CPA_i \quad (11)$$

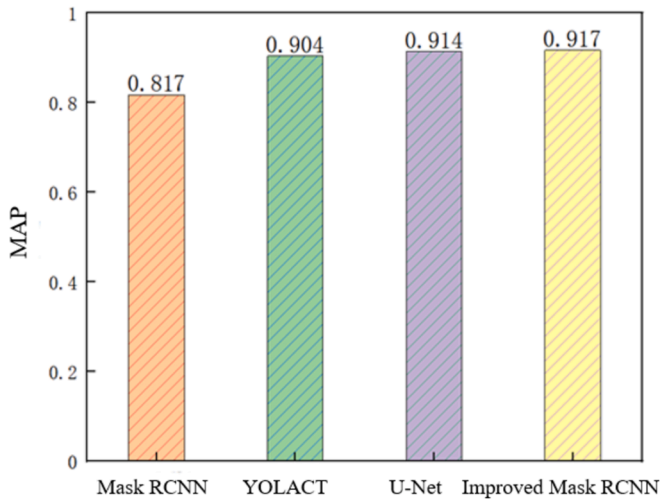
In this experiment, MIoU and mAP were mainly used as the evaluation indicators of the network segmentation effect. The confidence level indicates the degree of confidence that the split box contains the ore target and the confidence that the split box includes all the characteristics of the entire ore target. Because the confidence threshold setting is different, there will be different prediction effects, so the confidence threshold is set to 0.5, 0.75, and 0.85 for experimental comparison, and the corresponding evaluation metrics are MIoU50, MIoU75, MIoU85, mAP50, mAP75, mAP85. In addition, the average value of the three thresholds was used as the evaluation index, which was expressed as MIoU and mAP, respectively.

To verify the superiority of the improved Mask R-CNN segmentation network, the improved Mask R-CNN segmentation network were trained on the self-made ore dataset, and three common segmentation algorithms, Mask R-CNN, YULAC, and U-Net, were evaluated under the same experimental environment and hyperparameters, respectively, and

Table 4

Test results for 4 networks mAP50, mAP75, mAP85 and mAP.

Network model	mAP50	mAP75	mAP85	mAP
Mask R-CNN	0.826	0.795	0.811	0.817
YOLOACT	0.912	0.893	0.908	0.904
U-Net	0.933	0.915	0.894	0.914
Improved Mask R-CNN	0.927	0.908	0.916	0.917

**Fig. 11.** MAP test results for 4 types of networks.

compared with the proposed improved Mask R-CNN segmentation network, and the comparison results are as follows.

As can be seen from Fig. 10 above, the segmentation effect of Mask R-CNN is the worst, and the segmentation effect of the improved algorithm is better, which is about 8.9 % higher than that of Mask R-CNN, which shows that the proposed improved segmentation algorithm has a significant improvement in the segmentation effect.

The following Table 4 and Fig. 11 compare the mAP test results of the four networks, and Fig. 11 shows the bar charts comparing the mAP test results of the four networks. As can be seen from the figure, among the four network structures, the Mask R-CNN has the worst segmentation effect, with a mAP of only 81.7%, while the mAP of the other three networks is above 90%, and the mAP value of the improved Mask R-CNN segmentation algorithm is 10% higher than that of the original algorithm.

By comparing the test results of the four network structures under the

MIOU and mAP indicators, it can be concluded that the improved segmentation algorithm proposed in this paper has excellent performance in both indicators, indicating that the proposed method has brought great improvement to the overall segmentation performance of the network. To understand the superiority of the improved segmentation algorithm more intuitively, Fig. 12 shows a comparison of the results of a set of four networks on the raw ore test set. It can be found that the test results of the improved Mask R-CNN algorithm are closer to the true label of the original ore image.

To further evaluate the performance of the improved Mask R-CNN model in handling the segmentation task of ore images under complex working conditions, this study presents the predicted results of segmentation of ore images in real crushing operation scenarios by both the Mask R-CNN and the improved Mask R-CNN models. As can be seen from the results in Figs. 2-15, the original Mask R-CNN has more missed detection issues in ore segmentation, while the improved model has reduced the phenomenon of missed detections to some extent, and the segmentation of ore edges is clearer. This result indicates that the segmentation effect of the improved model has been enhanced in actual working scenarios.

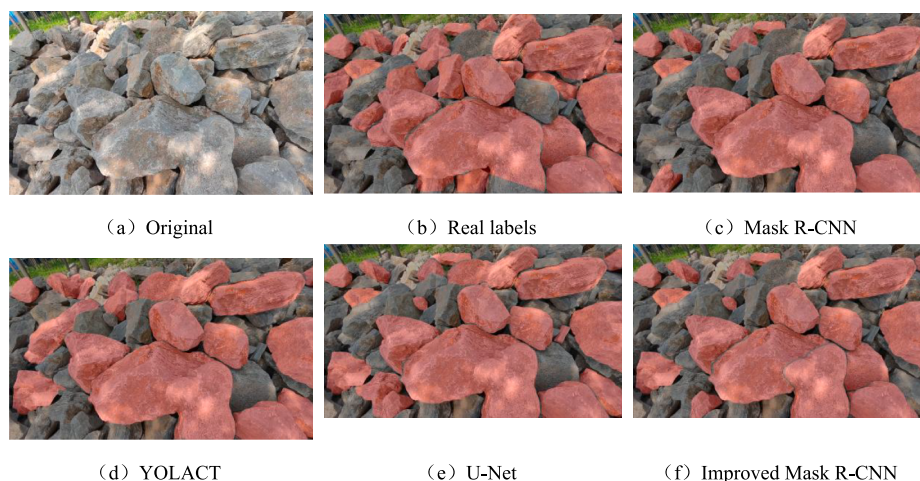
4.3. Accuracy of ore target positioning based on sensor fusion

To evaluate the positioning accuracy of the above method, the position of a pile of ore is determined as shown in the Fig. 14 below in the front environment, the gray and black sector areas represent the horizontal detection area and the vertical detection area of the lidar respectively, and the three-dimensional coordinates of the center point of the upper surface of the ore target are measured, and the experimental results under the three conditions of actual position, industrial camera detection, and fusion scheme detection are compared, as shown in the following table.

From the test results in the above Table 5, it can be seen that the average error of the X-axis is 5.07mm, the average error of the Y-axis and the average error of the Z-axis of the industrial camera is 5.07mm, the average error of the Y-axis is 2.9mm, and the average error of the Z-axis is 4.03mm, and the average error of the X-axis, Y-axis, and Z-axis is 3.03mm. Compared with the detection results of industrial cameras, the detection accuracy of the industrial camera and lidar fusion scheme in three dimensions is significantly improved.

5. Conclusion

To address the detection and positioning of multiple ore targets in complex scenarios after secondary blasting, traditional recognition

**Fig. 12.** Comparison of the results of the four networks on the raw ore segmentation.

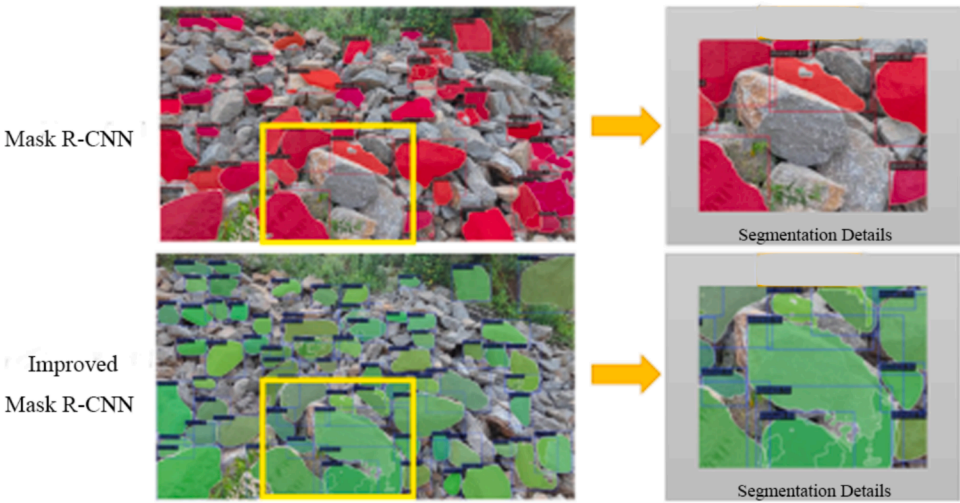


Fig. 13. Comparison of Ore Segmentation Between Improved and Original Algorithms in Real-World Operating Scenarios.

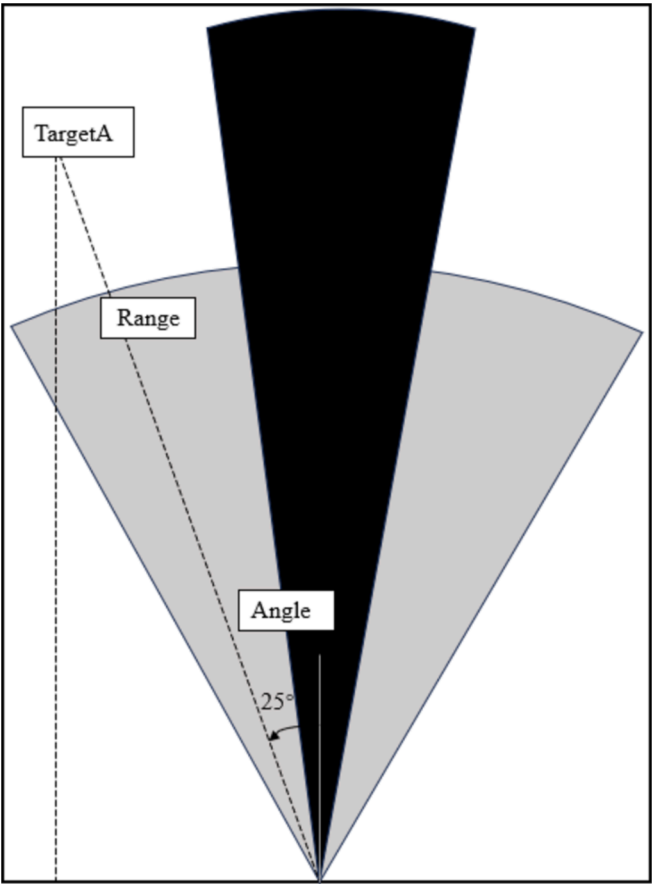


Fig. 14. Radar measurement parameters.

methods fail to meet the demands of multi-target image segmentation tasks. To achieve better segmentation results, an improved model based on Mask-RCNN is proposed for precise ore detection and segmentation. This includes an improved overall framework, as well as enhancements to the backbone network and FPN tailored to the characteristics of the ore. The model’s performance was evaluated through experiments on a self-built dataset. Additionally, the improved Mask-RCNN target segmentation algorithm extracts pixel information from ore images, which is then fused with the detected three-dimensional coordinates of the

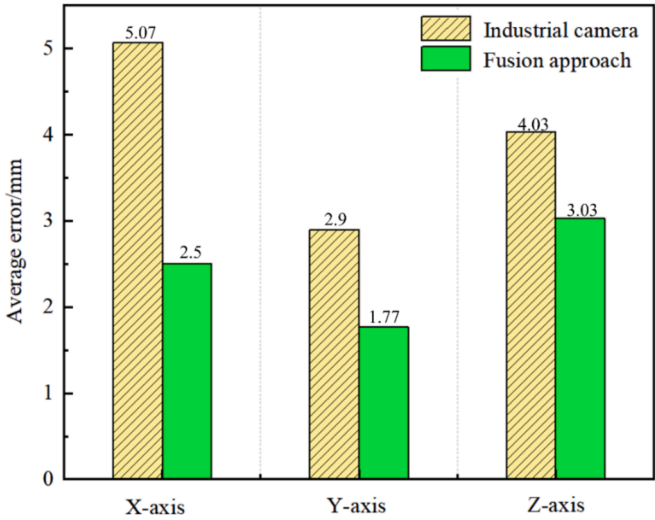


Fig. 15. Comparison chart of 3D positioning test error.

ores. Using the positioning principles of industrial cameras, spatial localization of ore targets is achieved. Experimental analysis indicates that the proposed method achieves higher detection accuracy, with mAP and MIoU improvements of 8.9% and 10.0%, respectively, compared to the original algorithm. The method also achieves average 3D positioning errors of 2.5mm, 1.77mm, and 3.03mm on the X, Y, and Z axes, respectively, which are lower than the average errors of 5.07mm, 4.03mm, and 4.03mm. These results demonstrate that the method effectively identifies ore targets in complex scenarios as expected and provides reliable ore location information for the hydraulic crusher perception system. Future work should focus on further improving ore recognition capabilities, reducing false recognition rates, and enhancing anti-interference performance in complex environments.

CRedit authorship contribution statement

Qihua Ma: Writing – review & editing, Supervision, Conceptualization. **Wenchao Wang:** Writing – original draft, Software, Data curation. **Xin Weng:** Writing – original draft, Investigation. **Qilin Li:** Visualization, Formal analysis.

Table 5

Results of the three-dimensional Positioning Test.

serial number	Actual coordinates /mm			Industrial cameras detect coordinates /mm			The fusion scheme detects the coordinates /mm		
	X	Y	Z	X	Y	Z	X	Y	Z
1	1237.2	170.5	161.8	1241.8	172.2	168.3	1239.1	169.3	165.2
2	1281.2	−121.1	181.9	1275.3	−106.8	176.4	1278.6	−108.3	178.6
3	970.5	20.2	80.7	975.2	22.9	80.6	972.5	18.9	78.3

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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